

Rehabilitation Exercises for Dysfunction of the Scapula

Exploration of Muscle Activity Using Fine-Wire EMG

Kelly R. Berckmans,^{*†} PT, Birgit Castelein,[†] PT, PhD, Dorien Borms,[†] PT, PhD, Thierry Parlevliet,[‡] MD, and Ann Cools,[†] PT, PhD
Investigation performed at Ghent University, Ghent, Belgium

Background: Scapular muscle activity during shoulder exercises has been explored with surface electromyography (EMG). However, knowledge about the activity of deeper-layer scapular muscles is still limited.

Purpose: To investigate EMG activation of the deeper-layer scapular stabilizers (levator scapulae [LS], rhomboid major [RM], pectoralis minor [Pm] muscles) together with superficial muscle activity (upper [UT], middle [MT], and lower trapezius [LT] and serratus anterior [SA]) during 4 exercises often used for training scapular function. Based on the amplitude EMG of the deeper-layer muscles, scapular muscle activation ratios for the 4 exercises were calculated, hereby providing knowledge of the optimal muscle balance.

Study Design: Descriptive laboratory study.

Methods: A total of 26 healthy participants performed 4 shoulder exercises (side-lying external rotation [ER], side-lying forward flexion, prone horizontal abduction with ER, and prone extension) while simultaneously measuring scapular muscle activity. Intramuscular electrodes were used for the deeper layer, in contrast to surface electrodes for the superficial muscles. All data were normalized to percentage of maximal voluntary isometric contraction (%MVIC), and the activation ratios (the muscle activity of the deeper layer relative to the other muscles) were calculated. A 1-way analysis of variance with Bonferroni correction was applied for statistical analysis.

Results: Moderate activity was found in all exercises for the LS and RM (25%-45% MVIC). The Pm resulted in low activity during both side-lying exercises (13%-18% MVIC). Ratios involving LS or RM showed values >1 for all exercises (1.28-12.41) except for LS/MT, LS/LT, and LS/RM (0.85-0.98) during side-lying ER, and LS/MT, RM/MT and RM/LS (0.85-0.99) during side-lying forward flexion. Likewise, values <1 were found when MT (0.85) and LS (0.99) were involved with RM in the numerator during side-lying forward flexion. Ratios with Pm in the numerator showed values <1 , apart from the ratios with UT and SA in the denominator.

Conclusion: The study provides extended knowledge about the deeper-layer scapular muscle activity and related ratios during the 4 shoulder exercises mentioned here. Putting theory into practice, based on our results, we advise both side-lying exercises to be performed to strengthen LT and MT, even in case of hyperactivity of the Pm. However, the 4 exercises should be given carefully to patients with hyperactivity in the LS and/or RM.

Clinical Relevance: The findings of this study may assist clinical decision making in exercise selection for restoring scapular function.

Keywords: shoulder; physical therapy/rehabilitation; electromyography

The causative relationship between the scapula and shoulder disorders is under debate; nevertheless, no doubt exists about the importance of the scapula in normal shoulder function.^{23,30,31,34} The scapula plays different crucial roles, such as creating a stable base for the glenohumeral joint, performing rotational and translational movements along the thoracic wall, avoiding subacromial conflicts by elevating the acromion, providing a site for muscle attachment

for glenohumeral as well as scapulothoracic muscles, and creating a link in the kinetic chain between the trunk and upper extremities.^{22,24-26} Therefore, scapular training is an inherent part of a shoulder rehabilitation program.

The optimal activation balance of scapular muscles is often the target for patients with scapular dysfunction, as imbalance could be (one of) the cause(s) of their shoulder complaints. Previous research reported the activation patterns of superficial scapular muscles using surface electromyography (EMG).[§] The gained insights endorse a clinician's (exercise) choice within the treatment plan of

a shoulder patient. Consequently, an optimal activation of both agonist and antagonist muscles ensures proper joint biomechanics.^{5,29} So, the right choice of exercises in scapula-focused rehabilitation is of importance. For example, the study by Cools et al¹² proposed a set of 4 exercises (side-lying external rotation [ER], side-lying forward flexion, prone horizontal abduction with ER, and prone extension) to optimize balance within the trapezius muscle. They reported an effective activation of the middle trapezius (MT) and lower trapezius (LT) muscle with low activation of the upper trapezius (UT) muscle.^{12,14} Therefore, the exercises could be recommended for patients with shoulder complaints linked to an overactive UT and hypoactivity of the MT and/or LT.

More recently, the activation patterns of the deep scapular stabilizers, such as the levator scapulae (LS), rhomboid major (RM), and pectoralis minor (Pm) muscles, were explored with fine-wire EMG during rehabilitation exercises.^{2,7,8,10} Exploring the activation of these muscles gives a new dimension of information that may influence the clinicians' exercise choice. Moreover, knowledge about the optimal muscle activation ratio (the relative activity of one muscle in perspective to another, rather than absolute levels of activity) may be of importance for improving scapular muscle strength and balance. Studies focusing on balance ratio rather than absolute muscle activation levels are limited.^{11,12,15,19,20,27} In this view, scapular rehabilitation exercises could specifically be chosen with the aim to selectively activate the weaker muscle with an inhibition of the hyperactive muscle, thus potentially restoring the scapular balance ratio.¹³ In clinical practice, the hyperactivation of the Pm and LS is often seen and can cause abnormal positioning and movement of the scapula. Thus, choosing the exercises with minimal activation of those deeper-layer muscles together with higher activation of weaker muscles could be of interest in those patients. However, these muscles and other deeper-layer scapular stabilizers, such as the RM, have not yet been examined in the frequently used rehabilitation exercises as published by Cools et al.¹²

Therefore, the first purpose of this study was to investigate the EMG activation of the deeper-layer scapular stabilizers (LS, RM, and Pm) during 4 exercises that are often used for training scapular function.¹² A second aim was to calculate scapular muscle activation ratios including the activity of the deeper-layer muscles. Ratios give information about the activity of one muscle compared with another during the exercises and may help the clinician in the choice of exercises. We hypothesized that there would be differences in EMG amplitude, as well as in ratios between the exercises.

METHODS

Participants

A total of 26 healthy individuals (mean age, 32 ± 12 years; mean height, 174.5 ± 7.75 cm; mean weight, 66.8 ± 8.81 kg; mean body mass index, 21.8 ± 1.78) participated voluntarily in this study. Participants were included if they were free from current or past shoulder and/or neck pain for the past 6 months and showed no loss of shoulder range of motion. Individuals were excluded if they performed upper limb strength training or overhead sports for >6 hours a week. All participants signed a written consent before testing. This study was approved by the ethical committee of the Ghent University Hospital.

General Design

The EMG activity of 7 scapulothoracic muscles (UT, MT, LT, serratus anterior [SA], LS, RM, and Pm) was measured during 4 different shoulder rehabilitation exercises.¹² The exercises are described in Table 1.

Test Procedure

The preparation for the EMG measurements consisted of shaving, scrubbing, and cleaning the participant's skin with alcohol to reduce impedance (<10 k Ω). A medical doctor, a specialist in fine-wire EMG (T.P.), inserted with ultrasound guidance and by use of a single-use 25-gauge hypodermic needle, the intramuscular paired hooked fine-wire electrodes (wire length, 125 mm; stainless steel; insulated nickel alloy wire; first wire stripped 2 mm, second wire insulated for 3 mm and then stripped 2 mm; Dickinson and Company) into the muscle belly of the 3 muscles (LS, RM, and Pm) according to the locations described by Delagi et al.¹⁶ For the other 4 scapulothoracic muscles (UT, MT, LT, and SA), surface electrodes (Ag/Ag Cl, type N-00-S; Ambu Blue Sensor P; 30×22 mm; Ambu A/S) were placed with a 1-cm interelectrode distance according to the instructions of Basmajian and De Luca.¹ A reference electrode was placed over the spinous process of the C7 vertebra. Both surface and fine-wire electrodes were looped and taped to the skin to avoid moving artifacts or being accidentally removed during the exercises. The EMG data were collected by a TeleMyo 2400 G2 Telemetry System (Noraxon USA Inc) with a sampling rate of 3000 Hz. The device had a common-mode rejection ratio of 100 dB. Gain was set at 1000 baseline noise <1 μ V root mean square.

*Address correspondence to Kelly R. Berckmans, PT, Department of Rehabilitation Sciences and Physiotherapy, Faculty of Medicine and Health Sciences, Ghent University, Corneel Heymanslaan 10, 3B3 ingang 46, Ghent, 9000, Belgium (email: kelly.berckmans@ugent.be).

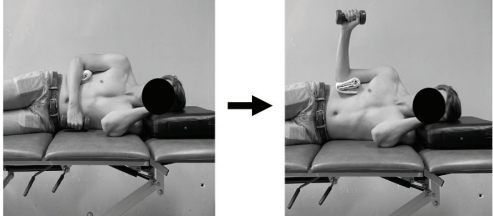
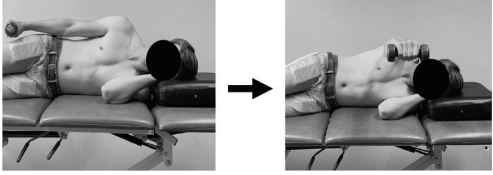
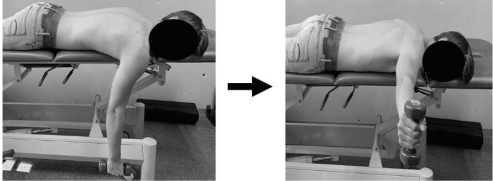
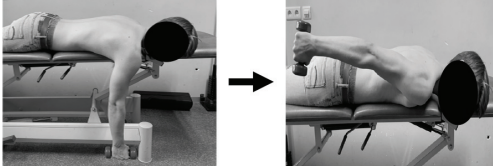
[†]Department of Rehabilitation Sciences and Physiotherapy, Faculty of Medicine and Health Sciences, Ghent University, Ghent, Belgium.

[‡]Physical Medicine and Orthopedic Surgery, Faculty of Medicine, University Hospital Ghent, Ghent, Belgium.

Submitted July 16, 2020; accepted March 10, 2021.

The authors declared that they have no conflicts of interest in the authorship and publication of this contribution. AOSM checks author disclosures against the Open Payments Database (OPD). AOSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

TABLE 1
Exercise Description^a

Exercises	Description	
Side-lying ER		Side-lying position with shoulder in neutral position and elbow flexed 90°. An ER was performed with a towel between elbow and trunk to avoid compensatory movements, starting with hand on the abdomen.
Side-lying forward flexion		Side-lying position with shoulder along the body. A 90° forward flexion was performed in the sagittal plane.
Prone horizontal abduction with ER		Prone position with shoulder resting in 90° forward flexion. A horizontal abduction to a horizontal position was performed, with ER of the shoulder (thumbs up) at the end of the movement.
Prone extension		Prone position with shoulder resting in 90° forward flexion. An extension to neutral position was performed with the shoulder in neutral rotation.

^aER, external rotation.

A short warm-up consisting of multidirectional shoulder movements was followed by the performance of maximal voluntary isometric contractions (MVIC). The protocol of Castelein et al⁹ was used, consisting of a set of 5 different isometric MVIC test positions allowing the EMG data to be normalized. The test position consisted of the following: (1) shoulder flexion to 135° (sitting), (2) arm abduction at 90° (sitting), (3) horizontal abduction with ER (prone lying), (4) arm raised above head in line with LT muscle fibers (prone lying), and (5) arm raised above head in line with Pm muscle fibers (supine lying). Each MVIC position was performed 3 times (5 seconds during contraction, controlled by metronome) with at least 30 seconds of rest between repetitions. A rest period of at least 90 seconds was applied between the different test positions. The MVIC positions as well as the applied manual pressure were taught to each individual by the same investigator (B.C.), who gave standardized encouragement during the MVIC to promote maximal effort.

Before measurements, the exercises were explained to the participants. Two test trials of each exercise were performed, and the movement was corrected if necessary. Then the participant performed 5 repetitions of the 4 exercises under supervision of the researcher. All exercises were performed on the dominant side only and in a random

order with dumbbells. The amount of weight used during the exercises was determined by sex and bodyweight based on a previous study (Table 2).¹⁴ Each exercise consisted of a 3-second concentric phase and 3-second eccentric phase. The speed of the movement was controlled by the use of a metronome (set at 60 beats/minute). A rest period of 5 seconds was taken between each trial. Between each exercise, a break of 90 seconds was applied. The Pm was measured only in the 2 side-lying exercises, as no contact with hardware material was allowed during the exercises in the prone position.

Signal Processing and Data Analysis

The MyoResearch 3.4 Master edition software (Noraxon USA Inc) was used for signal processing. Cardiac artifacts were subsequently removed from the signals, followed by a 20-Hz high-pass Butterworth filter, a full-wave rectification, and smoothing (root mean square of 100 ms). The mean amplitude for each muscle and each participant was computed for each exercise across the 3 intermediate repetitions of the completed 5 repetitions. The first and last repetitions were not used to prevent the data being influenced by habituation or fatigue. All EMG activity

TABLE 2
Weights Used by the Participants for the 4 Exercises^{14a}

Sex/Body Mass, kg	Side-Lying ER	Side-Lying Forward Flexion	Prone Horizontal Abduction With ER	Prone Extension
Male				
50-59	2.5	2.5	1.5	1.5
60-69	3.0	3.0	2.0	2.0
70-85	3.5	3.5	3.0	3.0
Female				
50-55	2.5	2.0	1.0	1.5
56-64	3.0	2.5	1.5	1.5
65-75	3.0	3.0	1.5	1.5

^aValues for the handheld weights are in kg. ER, external rotation.

values were normalized and expressed as a percentage of MVIC (%MVIC).

For each MVIC, the average EMG amplitude was calculated over a 2.5-second window of the 5 seconds. The mean of the 3 trials was used for normalization. For all muscles except the Pm, the highest mean EMG value over all 4 positions was taken, determining the MVIC used for the normalization of the EMG activity of the exercises. For the Pm, the EMG value of the supine position with the arm raised above the head in line with Pm muscle fibers was used.

Statistical Analysis

The normalized EMG activity (%MVIC) for all muscles was computed by dividing the raw EMG activity during the exercises expressed in microvolts (μ V) by the MVIC for each specific muscle, multiplied by 100. The activation ratio of the deeper-layer muscles (LS, RM, and Pm) relative to the other muscles was calculated. The ratio between 2 superficial muscles (trapezius and SA) or muscle parts (trapezius) was not computed as these were previously described by Cools et al.¹² The statistical software SPSS Version 26.0 (IBM corp) was used for all statistical analyses.

Means and standard deviations were calculated for the normalized EMG activity for all muscles investigated during all 4 exercises. However, for the Pm, EMG activity was computed only for both side-lying exercises because the prone position was not allowed due to possible contact with hardware material. A 1-way analysis of variance (ANOVA) was applied to determine if there were significant differences in EMG activity and muscle ratios between "exercises" (side-lying ER, side-lying forward flexion, prone horizontal abduction with ER, prone extension) for each muscle. Post hoc pairwise comparisons were performed using a Bonferroni correction for all muscles except the Pm. Because the Pm was measured only during 2 exercises, no post hoc tests were needed for this muscle. The significance level was set at an alpha level of .05.

To facilitate the interpretation, the values of EMG were categorized as low activity (<20% MVIC), moderate activity (20%-50% MVIC), or high activity (>50% MVIC).³

RESULTS

EMG Activity During the 4 Exercises

The descriptive analysis and results of the 1-way ANOVA are presented in Table 3.

In general, moderate activity was found for the LS and RM in all exercises and low activity of the Pm in both side-lying exercises. The LS has remarkable lower activation in the side-lying ER and prone extension (25%-28% MVIC) compared with the other 2 exercises (42%-45% MVIC).

Muscle Balance Ratios Based on the EMG Activity During the 4 Exercises

The results for the ratios are reported in Table 4.

For the interpretation of the results, a ratio >1 reflects higher muscle activity in the numerator compared with denominator; the opposite is a ratio <1.

For the ratios involving the LS or RM in the numerator, almost all values were >1 for all exercises, except for LS/MT, LS/LT, and LS/RM (0.85-0.98) approaching 1 for the side-lying ER, and LS/MT (0.99) for the side-lying forward flexion exercise. The RM revealed a likewise exception for the RM/MT and RM/LS (0.85-0.99). Concerning the ratios with Pm in the numerator, results showed values <1 with MT, LT, LS, or RM in the denominator and >1 with UT or SA in the denominator.

DISCUSSION

The aim of this study was to provide new insights into EMG activity of the scapular muscles during the exercises intended to restore the scapular muscle balance, exercises that were previously published by Cools et al.¹² The innovative aspect was the additional use of fine-wire EMG to explore the activity of the deeper-layer scapular muscles, namely the LS, RM, and Pm. This allowed us to provide relevant muscle balance ratios between the deep and superficial scapular muscles during these exercises.

The activity of both the LS (25%-45% MVIC) and RM (28%-42% MVIC) was moderate during the 4 exercises.

TABLE 3
Details of the Comparative Statistical Analysis for EMG Measurements Based on Exercises^a

Muscle	SER	SFF	PHABER	PEX	P Value	P Value	
					1-Way ANOVA	Pairwise Comparisons After Bonferroni Correction	
UT	5.41	16.31	15.21	5.90	<.001 ^b	SER-SFF: <.001 ^b	SFF-PHABER: ≥.999
	[4.24-6.57]	[13.61-19.01]	[12.93-17.49]	[4.48-7.32]		SER-PHABER: <.001 ^b	SFF-PEX: <.001 ^b
MT	35.64	50.69	38.57	17.99	<.001 ^b	SER-PEX: ≥.999	PHABER-PEX: <.001 ^b
	[28.22-43.06]	[40.54-60.85]	[33.08-44.06]	[14.75-21.23]		SER-SFF: .015 ^b	SFF-PHABER: .083
LT	31.22	35.60	38.33	24.32	.042 ^b	SER-PHABER: ≥.999	SFF-PEX: <.001 ^b
	[23.80-38.64]	[29.68-41.53]	[32.79-43.87]	[14.17-34.47]		SER-PEX: .003 ^b	PHABER-PEX: <.001 ^b
SA	8.97	14.83	7.04	12.27	.010 ^b	SER-SFF: ≥.999	SFF-PHABER: ≥.999
	[5.89-12.06]	[10.11-19.55]	[4.79-9.28]	[8.54-16.01]		SER-PHABER: ≥.999	SFF-PEX: .182
LS	24.64	42.04	45.28	28.49	≤.001 ^b	SER-PEX: ≥.999	PHABER-PEX: .045 ^b
	[19.26-30.03]	[33.71-50.38]	[34.14-56.41]	[20.60-36.39]		SER-SFF: .111	SFF-PHABER: .012 ^b
RM	32.57	39.09	41.96	27.58	.072	SER-PHABER: ≥.999	SFF-PEX: ≥.999
	[25.04-40.10]	[29.94-48.23]	[32.85-51.08]	[18.82-36.34]		SER-SFF: .021 ^b	PHABER-PEX: .026 ^b
Pm	12.76	17.68			.154	SER-PEX: ≥.999	SFF-PHABER: ≥.999
	[7.48-18.04]	[13.07-22.29]				SER-PHABER: .711	SFF-PEX: .325
						SER-PEX: ≥.999	PHABER-PEX: .100

^aData are reported as mean [95% CI]. Values are expressed in percentage of maximal voluntary isometric contractions (%MVIC): low <20 %MVIC (white cells), moderate 20-50 %MVIC (medium gray cells), high >50 %MVIC (dark gray cells). ANOVA, analysis of variance; EMG, electromyography; LS, levator scapulae; LT, lower trapezius; MT, middle trapezius; PEX, prone extension; PHABER, prone horizontal abduction with external rotation; Pm, pectoralis minor; RM, rhomboid major; SA, serratus anterior; SER, side-lying external rotation; SFF, side-lying forward flexion; UT, upper trapezius.

^bP < .05 indicates statistical significance.

On the other hand, the anteriorly situated Pm muscle showed low activity (13%-18% MVIC) during the 2 exercises in which the Pm was measured.

Knowledge about the deeper-layer muscle activity made it possible to calculate the muscle balance ratios, including these muscles. The ratios were in the range of 0.85 to 12.41 with the LS in the numerator. Similar results with slightly higher ratios were found for the RM (0.85-10.65). Overall, both muscles were more active than the others (denominator muscles) during the exercises. Also, the 95% CIs provide information to take into consideration when interpreting the results.

A differentiation between ratios above and below 1 should be made. The 95% CIs for the Pm/MT (0.28-0.56/0.30-0.59), Pm/LT (0.30-0.57/0.38-0.76), Pm/LS (0.32-0.71/0.31-0.56), and Pm/RM (0.31-0.58/0.35-0.84), respectively are all <1 during the side-lying exercises. In conclusion, the Pm—acting as numerator—shows low activation compared with the denominator. All 95% CIs >1 revealed moderate or high activation of the numerator muscle, resulting in a relative inhibition of the denominator muscle. As those intervals are extended, looking at absolute values is of importance. Regarding the 95% CIs that start below and end above 1, as seen for the RM, the results show a mean value of nearly 1. The RM seems to be alternatively more or less active compared with the muscle in the denominator. In other words, during some exercises the activation pattern of the RM differs between individuals.

In rehabilitation, the selection of the appropriate exercises should be custom-made and based on proper clinical reasoning. When higher activation of the LS and RM is preferred in rehabilitation, these exercises would be suitable. However, LS hyperactivity has been suggested in patients with chronic neck pain and shoulder girdle disorders.^{4,6} On the other hand, the RM acts as a scapular elevator and downward rotator. An overactive RM could depress the glenoid cavity with subacromial pain syndrome as a result. In conclusion, if rehabilitation aims for a lower LS and RM activity, these exercises should only be prescribed with caution and careful observation of movement performance.

Concerning the Pm, results revealed ratios <1 (0.42-0.60) for both exercises, which means that the Pm is less active than the other explored muscles during the side-lying exercises. In contrast, the opposite was found for the Pm/UT and Pm/SA ratio (1.23-2.93), where the Pm was slightly more active compared with the UT or SA during the side-lying exercises, with a ratio >1 as the outcome. However, the ratio being >1 as described for Pm/UT and Pm/SA is, in the end, not clinically relevant. Because both the mean difference and the muscle activity of all involved muscles are low, interpretation of the ratio should be made with some caution. Both the ratio and amplitude value have to be taken into account during the choice of exercises.

Putting theory into practice, based on our results we advise both side-lying exercises be performed to strengthen

TABLE 4
Detail of the Comparative Statistical Analysis for EMG Measurements Based on Exercises

Muscle	SER	SFF	PHABER	PEX	<i>P</i> Value	<i>P</i> Value	
					1-way ANOVA	Pairwise Comparisons After Bonferroni Correction	
LS/UT	6.04	3.23	3.25	6.63	.004 ^b	SER-SFF: .114	SFF-PHABER: ≥.999
	[4.12-7.96]	[2.26-4.20]	[2.36-4.15]	[4.08-9.18]		SER-PHABER: .112	SFF-PEX: .031 ^b
LS/MT	0.85	0.99	1.35	1.72	≤.001 ^b	SER-PEX: ≥.999	PHABER-PEX: .030 ^b
	[0.65-1.06]	[0.76-1.21]	[0.98-1.72]	[1.29-2.15]		SER-SFF: ≥.999	SFF-PHABER: .633
LS/LT	0.98	1.43	1.37	1.93	.054	SER-PHABER: .149	SFF-PEX: .009 ^b
	[0.72-1.24]	[1.00-1.85]	[1.00-1.75]	[1.15-2.70]		SER-PEX: <.001 ^b	PHABER-PEX: .577
LS/SA	6.53	4.79	12.41	4.49	.003 ^b	SER-SFF: ≥.999	SFF-PHABER: ≥.999
	[3.31-9.74]	[2.98-6.60]	[6.94-17.88]	[2.63-6.35]		SER-PHABER: ≥.999	SFF-PEX: .873
LS/RM	0.96	1.41	1.28	1.72	.028 ^b	SER-PEX: .037 ^b	PHABER-PEX: .620
	[0.73-1.18]	[1.08-1.74]	[1.02-1.54]	[1.16-2.29]		SER-SFF: ≥.999	SFF-PHABER: .011 ^b
LS/Pm	3.67	3.93			.770	SER-PHABER: .085	SFF-PEX: ≥.999
	[2.33-5.01]	[2.67-5.20]				SER-PEX: ≥.999	PHABER-PEX: .007 ^b
RM/UT	7.37	2.75	3.22	5.97	≤.001 ^b	SER-SFF: .466	SFF-PHABER: ≥.999
	[5.35-9.39]	[2.01-3.50]	[1.92-4.52]	[3.21-8.73]		SER-PHABER: ≥.999	SFF-PEX: ≥.999
RM/MT	1.07	0.85	1.17	1.73	.013 ^b	SER-PEX: .019 ^b	PHABER-PEX: .499
	[0.66-1.48]	[0.66-1.04]	[0.90-1.44]	[1.14-2.32]		SER-SFF: .003 ^b	SFF-PHABER: ≥.999
RM/LT	1.24	1.26	1.17	2.04	.054	SER-PHABER: .010 ^b	SFF-PEX: .082
	[0.88-1.60]	[0.81-1.71]	[0.94-1.41]	[1.21-2.87]		SER-PEX: ≥.999	PHABER-PEX: .204
RM/SA	7.29	3.91	10.65	3.61	≤.001 ^b	SER-SFF: ≥.999	SFF-PHABER: ≥.999
	[4.66-9.92]	[2.72-5.10]	[6.10-15.19]	[2.02-5.20]		SER-PHABER: ≥.999	SFF-PEX: .011 ^b
RM/LS	1.48	0.99	1.03	1.25	.202	SER-PEX: .106	PHABER-PEX: .251
	[1.07-1.88]	[0.68-1.30]	[0.75-1.30]	[0.79-1.71]		SER-SFF: ≥.999	SFF-PHABER: ≥.999
RM/Pm	5.53	3.44			.361	SER-PHABER: ≥.999	SFF-PEX: .193
	[1.24-9.81]	[1.76-5.12]				SER-PEX: .166	PHABER-PEX: .106
Pm/UT	2.76	1.23			.009 ^b	SER-SFF: .484	SFF-PHABER: .004 ^b
	[1.66-3.86]	[0.89-1.56]				SER-PHABER: .501	SFF-PEX: ≥.999
Pm/MT	0.42	0.45			.764	SER-PEX: .333	PHABER-PEX: .002 ^b
	[0.28-0.56]	[0.30-0.59]				SER-SFF: .366	SFF-PHABER: ≥.999
Pm/LT	0.44	0.57			.238	SER-PHABER: .472	SFF-PEX: ≥.999
	[0.30-0.57]	[0.38-0.76]				SER-PEX: ≥.999	PHABER-PEX: ≥.999
Pm/SA	2.93	1.95			.177		
	[1.64-4.22]	[1.22-2.67]					
Pm/LS	0.52	0.43			.460		
	[0.32-0.71]	[0.31-0.56]					
Pm/RM	0.44	0.60			.264		
	[0.31-0.58]	[0.35-0.84]					

^aData are reported as mean [95% CI]. ANOVA, analysis of variance; EMG, electromyography; LS, levator scapulae; LT, lower trapezius; MT, middle trapezius; PEX, prone extension; PHABER, prone horizontal abduction with external rotation; Pm, pectoralis minor; RM, rhomboid major; SA, serratus anterior; SER, side-lying external rotation; SFF, side-lying forward flexion; UT, upper trapezius.

^b $P < .05$ indicates statistical significance.

the LT and MT, as originally described,¹² even in case of hyperactivity of the Pm. However, the 4 exercises should be given carefully to patients with hyperactivity in the LS and/or RM. Observation of the movement during the exercises is highly recommended, looking for compensatory

movements due to hyperactivity of these muscles during the patients' performance.

Concerning the superficial muscles, low activity was found for the UT (5%-16% MVIC) and SA (9%-15% MVIC) contrary to the MT (18%-51% MVIC) and LT

(24%-38% MVIC), which were moderately active during the exercises. Although the activity of a number of muscles during the exercises is similar, as reported by Cools et al,¹² a general observation between results revealed lower values in our study. The differences between the results may be explained by the different protocol used in the two studies for the performance of the MVIC. For the present study, the recently published reliable protocol by Castelein et al⁹ was used, in which the highest value of activity during different testing positions was used.

A major strength of this study was that, measurements of the deeper-layer muscle activity were performed during popular shoulder rehabilitation exercises. Nevertheless, some limitations should be taken into account. The use of fine-wire EMG made it possible to explore the deeper layer of the scapular muscles. However, as this technique might be slightly painful, participants could have encountered some discomfort. Discomfort might have hindered the production of maximal activity during the performance of the MVICs. In view of minimizing those inconvenient factors, a specialized medical doctor performed all insertions of the fine wires. Also, some precautions were taken to minimize traction on the wires by use of a loop before adhering them to the skin.

Another limitation may be found in the absence of 3-dimensional kinematic analysis of scapular movements. The 3-dimensional kinematic analysis could have given additional relevant information but was not performed because of technical limitations, mainly as a result of the individuals' side-lying or prone position during the exercises. In the recent study by Berckmans et al,² those additional measurements revealed new insights about scapular position during commonly used shoulder rehabilitation exercises together with the activity of the deeper-layer muscles with individuals in an upright position.

The inclusion of only healthy individuals in this study can be seen as a potential limitation, but this was done because EMG results are a baseline for the prescription of exercises. However, obtaining knowledge of the activity in the deeper-layer muscle in the painful shoulder (upper extremity disorders) and in a population suffering from neck pain could be interesting for further research.

CONCLUSION

The aim of this study was to provide extended knowledge about the electromyographic activity in the deeper-layer scapular muscles during 4 commonly used rehabilitation exercises. In general, moderate activity was found for the LS and RM, in contrast with low activity for the Pm during the selected exercises. Overall, the ratios were >1 when the LS and RM were involved, in contrast to lower ratios when the Pm was taken into account. The findings of this study may assist the decision making of a clinician in devising a rehabilitation program. Putting theory into practice, based on our results, we advise both side-lying exercises to be performed to strengthen LT and MT, even in case of hyperactivity of the Pm. However, the 4 exercises

should be given carefully to patients with hyperactivity in the LS and/or RM.

ACKNOWLEDGMENT

The authors thank the master's students who helped during the measurements and all the volunteers who participated in this study.

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